SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 50%

Duration : 1 Hour
QUESTIONS: 5 **Total Marks:50**

Annexure A

Technical Presentation

Code of Conduct:

I Hemalakshmi H – 192571049 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Technical Presentation

1. Reactive Agent-Based Traffic Control System
Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis
Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations
Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System
Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management
Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

1. Reactive Agent-Based Traffic Control System

Overview

A Reactive Agent is an AI entity that responds instantly to environmental changes without long-term planning. In traffic control, this agent continuously senses traffic flow and adjusts signals to minimize congestion.

System Motivation

Urban traffic congestion causes:

- Increased travel time
- Higher fuel consumption
- Emergency vehicle delays
- Pollution spikes

A reactive agent solves this by **responding immediately** to real-time traffic conditions.

Architecture

1. Data Acquisition Layer

- IoT sensors (vehicle count, speed sensors)
- CCTV cameras with object detection
- GPS data from connected vehicles
- Roadside units (RSUs)

2. Streaming Data Processing

- Spark Streaming / Kafka pipelines
- Real-time metrics:
    - Queue length per lane
    - Average waiting time
    - Vehicle density
    - Emergency vehicle detection

3. Reactive Agent Logic

Perception: Reads current traffic state every few seconds.

Decision Rules:

- If density > threshold → extend green signal
- If queue length is low → shorten green
- If emergency vehicle detected → override normal cycle
- If pedestrian request triggered → allocate crossing time

Action: Updates traffic signal phases dynamically.

4. Expected Outcomes

- Reduced congestion
- Faster lane clearance
- Priority for emergency vehicles
- Adaptive signal timing

2. Goal-Based Agent for Healthcare Diagnosis

Overview

A Goal-Based Agent works by setting goals and choosing actions that achieve those goals. In healthcare, the agent aims to diagnose diseases and recommend optimal treatments.

System Motivation

Healthcare diagnosis requires:

- Accurate interpretation of patient data
- Fast decision-making
- Personalized treatment plans

A goal-based agent ensures structured, goal-driven reasoning.

Architecture

1. Data Pipeline

- Patient history
- Lab test results
- Medical imaging
- Vital signs
- Wearable device data

Data is cleaned, normalized, and stored in a clinical data warehouse.

2. Goal Formulation

Primary goals:

- **G1:** Identify the most probable disease
- **G2:** Recommend the best treatment plan
- **G3:** Minimize risk and side effects

3. Decision-Making Logic

- Rule-based reasoning (clinical guidelines)
- Decision trees for symptom analysis
- Probabilistic models (Bayesian networks)
- Risk scoring algorithms

4. Treatment Pathway Selection

Agent evaluates:

- Drug effectiveness
- Patient allergies
- Comorbidities
- Cost and availability

5. Expected Outcomes

- Faster diagnosis

- Evidence-based treatment
- Reduced medical errors
- Personalized care

3. Utility-Based Agent for E-Commerce Recommendations

Overview

A Utility-Based Agent selects actions that maximize a utility function. In e-commerce, it recommends products that maximize user satisfaction and business goals.

System Motivation

E-commerce platforms must:

- Personalize recommendations
- Increase conversion rates
- Improve user satisfaction

Utility-based agents optimize these outcomes mathematically.

Architecture

1. Data Sources

- User browsing history
- Purchase records
- Ratings and reviews
- Product metadata
- Price and discount information

2. Distributed Processing

Using PySpark:

- Compute relevance scores
- Analyze user behavior at scale
- Cluster similar users/items
- Calculate purchase likelihood

3. Utility Function

Utility combines:

- Relevance
- Satisfaction
- Price attractiveness
- Novelty
- Business revenue

Example:

$$U(i, temuser) = w_1 R + w_2 S + w_3 P + w_4 N$$

4. Agent Behavior

- Assigns utility to each product
- Ranks items based on utility
- Recommends top-N items
- Learns from feedback (clicks, purchases)
- Adjusts weights dynamically

5. Expected Outcomes

- Highly personalized recommendations
- Increased sales
- Better user engagement

4. Learning Agent for Personalized Education System

Overview

A Learning Agent improves its performance over time using feedback. In education, it adapts content difficulty and learning paths for each student.

System Motivation

Students have:

- Different learning speeds
- Different strengths and weaknesses
- Different content preferences

A learning agent ensures personalized education at scale.

Architecture

1. Data Collection

- Quiz scores
- Time spent on lessons
- Mistake patterns
- Content usage logs
- Engagement metrics

2. Distributed Processing (PySpark)

- Aggregate performance metrics
- Detect weak topics
- Identify learning patterns
- Predict dropout risk

3. Agent Components

- **Performance Element:** Chooses next content
- **Learning Element:** Updates internal model
- **Critic:** Evaluates outcomes
- **Feedback Loop:** Reinforces successful strategies

4. Adaptive Behavior

- Increases difficulty when student performs well
- Provides remedial content when struggling
- Recommends personalized learning paths
- Learns which teaching method works best

5. Expected Outcomes

- Improved learning outcomes
- Higher engagement
- Reduced dropout
- Personalized education for thousands of learners

5. Multi-Agent System for Disaster Management

Overview

A Multi-Agent System (MAS) uses multiple agents that collaborate to detect disasters and coordinate responses.

System Motivation

Disaster management requires:

- Fast detection
- Accurate risk assessment
- Coordinated response
- Real-time communication

MAS provides decentralized intelligence.

Agent Types

1. Sensor Agents

- Collect data from IoT sensors
- Monitor water levels, tremors, weather
- Analyze satellite images
- Track social media signals

2. Decision Agents

- Fuse data from multiple sensors
- Compute risk scores
- Predict disaster severity
- Trigger alerts

3. Communication Agents

- Send warnings to authorities
- Coordinate evacuation routes
- Share updates across regions

Coordination Strategies

- **Blackboard system:** Shared knowledge base
- **Message passing:** Agents communicate states
- **Distributed decision-making:** Local + global coordination
- **Real-time updates:** Continuous monitoring

Expected Outcomes

- Early disaster detection
- Faster response
- Reduced casualties
- Efficient resource allocation